Multimedia Signal Processing - 2nd Module

January, 16 2019

1. **Question (8 pts)**
   Consider the sequence of symbols emitted by a source characterized by an infinite elements alphabet

$$X = [14, 15, 18, 16, 22, 17, 18, 13, 15],$$

   and a quantizer with the following quantization rule

$$a_1 = -\infty,\ a_2 = -1,\ a_3 = 1,\ a_4 = +\infty,$$

$$Q(x) = I_k,\ \text{if } x \in (a_k, a_{k+1}]. \qquad I_k = \begin{cases} a_{k+1} - 1, & \text{if } k = 1, \\ \frac{a_k + a_{k+1}}{2}, & \text{if } k = 2, \\ a_k + 1, & \text{if } k = 3. \end{cases}$$

   (a) Encode X using closed-loop DPCM coding, thus obtaining $\tilde{X}$ (assume that the first symbol is known by the encoder, and use the previous symbol as predictor).

   (b) Compute the MSE introduced on $\tilde{X}$.

   (c) Consider the scenario of further compress the sequence X using Huffman coding in addition to DPCM. Compute the Huffman code and the average amount of needed bit per symbol (hint: do not consider the first symbol).

   (d) Encode X using open-loop DPCM coding, thus obtaining $\hat{X}$ (assume that the first symbol is known by the encoder, and use the previous symbol as predictor).

   (e) Compute the MSE introduced on $\hat{X}$.

2. **Question (5 pts)**
   Explain the concept of visual irrelevancy and visual redundancy in image perceptual coding. Report examples and comment them.

3. **Question (5 pts)**
   Consider two frames of a video sequence. Assuming one frame as the reference, describe how to perform motion estimation and motion compensation.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider a source X emitting symbols from an infinite alphabet with probability distribution function $p(x)$.

   - [T] [F] Lloyd-Max quantizer can only be used to quantize the symbols if $p(x)$ is not uniform.
   - [T] [F] Lossless coding can be applied to encode symbols emitted by X, but lossy coding ensures better compression.
   - [T] [F] Coding by means of DPCM guarantees less distortion than simple PCM at the same bitrate.

   (b) Consider a transform coding scheme.

   - [T] [F] Turning input data into transform coefficients can be done by matrix multiplication with any real valued matrix.
   - [T] [F] KLT transform is data-dependent and must be computed on purpose on the data we want to encode.
   - [T] [F] DCT transform is a good approximation of KLT for coding natural images.
   - [T] [F] Using a KLT scheme is faster than using a DCT one.

   (c) Consider lossless coding of symbols emitted by a source Y characterized by a finite element alphabet.

   - [T] [F] It is not possible to obtain a distortion higher than σ_y^2 (i.e., the source variance).
   - [T] [F] It is not possible to encode these symbols with rate less than $H(Y)$ (i.e., the entropy of the source).
   - [T] [F] Given a fixed rate, there is only one possible achievable distortion according to rate-distortion theory.

5. **Question (Matlab - 10pts)**

   You have two audio files (`gb_l.wav` and `gb_r.wav`) representing the left and right channels of a song, respectively. Your goal is to encode the two tracks jointly by using transform coding and quantization.

   (a) Load both audio tracks and plot the two waveforms.

   (b) Trim the first 0.5 seconds of each track and discard the rest. How many samples are you keeping?

   (c) Concatenate the two tracks into a $N \times 2$ matrix, where N is the number of samples you are considering from point (b).

   (d) Encode the concatenated audio tracks by applying transform coding and quantization with the following parameters:

   - 2D-DCT transform on blocks of size 2×2;
   - Uniform scalar quantization with step size 0.01.

   (e) Compute the average amount of bits per symbol needed to store the encoded tracks (hint: use entropy as rate).

   (f) Decode the encoded tracks and obtain an estimate of the left and right audio channels.

   (g) Compute the introduced distortion in terms of signal-to-noise ratio in dB.